Epinephrine Infusion

This protocol is for Critical Care Paramedic use only

Purpose

To authorize initiation and titration of a pumped epinephrine infusion on emergency responses for patients with hypotension that persists despite appropriate fluid resuscitation and repeated push dose epinephrine.

Preconditions

All of the following are required before initiating an epinephrine infusion under this protocol.

1. An IV infusion pump is available and functioning. Epinephrine infusion must be delivered via an IV pump. Infusion by gravity, dial-a-flow, or manual drip rate is not permitted under this protocol.
2. The administering paramedic holds Critical Care Paramedic privileges under Medical Control Privileges (Section 8-17).
3. The administering paramedic has completed Medical Control Authority approved training in vasopressor infusion administration and remains current.
4. Patent IV or IO access is confirmed, with blood return or free flow verified immediately prior to initiation.
5. Continuous cardiac monitoring, continuous pulse oximetry, and non-invasive blood pressure monitoring are in place.

Indications

Hypotension persisting after appropriate fluid resuscitation, where push dose epinephrine is already indicated and repeated dosing is required or anticipated, in any of the following:

1. Shock (Section 1-5)
2. Anaphylaxis/Allergic Reaction (Section 1-6)
3. Crashing Adult/Impending Arrest (Section 3-7)
4. Bradycardia, persistent and symptomatic after atropine and pacing (Section 5-2)
5. Pulmonary Edema/Cardiogenic Shock (Section 5-4)
6. Return of Spontaneous Circulation (Section 5-6)
7. Pediatric Crashing Patient/Impending Arrest (Section 4-9)
8. Pediatric Bradycardia (Section 6-2)
9. Pediatric Return of Spontaneous Circulation (Section 6-4)

The referring protocol governs the clinical indication. This protocol governs preparation, initiation, titration, and monitoring of the infusion.

Exclusions

1. Hemorrhagic shock with uncontrolled or ongoing hemorrhage. Vasopressor support is not a substitute for hemorrhage control and volume resuscitation. Refer to Hemorrhagic Shock (Section 2-14).
2. Hypovolemia that has not been addressed with appropriate fluid resuscitation.
3. Cardiac arrest. Refer to the applicable cardiac arrest protocol.
4. No infusion pump available, or the administering paramedic does not hold Critical Care Paramedic privileges and current training under this protocol.

Preparation

Recommended concentration is 2 mcg/mL, prepared as below. Where a different concentration is prepared, the rate tables in this protocol do not apply and rates must be recalculated for the concentration in use.

1. Add 1 mg of epinephrine (1 mL of 1 mg/mL) to a 500 mL bag of normal saline.
2. Final concentration is 2 mcg/mL.
3. Apply a completed medication-added label to the bag showing drug, total dose, final concentration, time, and preparer.
4. Verify concentration and pump programming with a second provider when a second provider is available.

Adult Dosing (patients greater than 14 years of age)

1. Begin at 2 mcg/min.
2. Titrate in 2 mcg/min increments every 3 to 5 minutes to endpoint.
3. Titrate to SBP greater than 90 mm/Hg.
4. Usual range is 2 to 10 mcg/min. Contact Medical Control before exceeding 10 mcg/min.

Adult Dose	Pump Rate at 2 mcg/mL
2 mcg/min	60 mL/hr
4 mcg/min	120 mL/hr
6 mcg/min	180 mL/hr
8 mcg/min	240 mL/hr
10 mcg/min	300 mL/hr

Pediatric Dosing (patients 14 years of age or less)

MI MEDIC cards do not carry epinephrine infusion dosing. Use the dosing and table below. Do not refer to MI MEDIC cards for infusion rates.

1. Begin at 0.1 mcg/kg/min, not to exceed the adult starting dose of 2 mcg/min.
2. Titrate in 0.1 mcg/kg/min increments every 3 to 5 minutes to endpoint, with each increment not to exceed 2 mcg/min.
3. Titrate to SBP greater than 70 mmHG + (2 x age in years) up to 100 mmHg.
4. Contact Medical Control before exceeding 1 mcg/kg/min, or before exceeding the adult ceiling of 10 mcg/min, whichever is reached first.
5. For any patient at or above 20 kg, use the adult starting rate and adult ceiling.

Weight	Starting Rate at 2 mcg/mL	Maximum Rate Without Medical Control
3 kg	9 mL/hr	90 mL/hr
5 kg	15 mL/hr	150 mL/hr
7 kg	21 mL/hr	210 mL/hr
10 kg	30 mL/hr	300 mL/hr
15 kg	45 mL/hr	300 mL/hr
20 kg or greater	60 mL/hr	300 mL/hr

For a weight not listed: starting rate in mL/hr equals weight in kg multiplied by 3, to a maximum of 60 mL/hr. Maximum rate without Medical Control in mL/hr equals weight in kg multiplied by 30, to a maximum of 300 mL/hr.

Transition from Push Dose Epinephrine

1. Discontinue push dose epinephrine once the infusion is running and titrating.
2. Do not administer push dose epinephrine and an epinephrine infusion concurrently except as a single bridging dose while the infusion is being prepared.
3. Document the time the infusion was initiated and the time push dose administration was discontinued.

Monitoring

1. Non-invasive blood pressure cycled every 3 minutes at minimum.
2. Continuous cardiac monitoring and continuous pulse oximetry.
3. Waveform capnography where an advanced airway is in place, per End Tidal Carbon Dioxide Monitoring-Procedure Protocol.
4. Inspect the infusion site every 5 minutes and with every rate change for signs of extravasation.
5. Document rate, concentration, and patient response with every titration.

Extravasation Response

Signs include swelling, blanching, pallor, coolness, pain at the site, or loss of free flow.

1. Stop the infusion immediately.
2. Aspirate residual medication through the catheter before removal, if able.

3. Remove the catheter.
4. Elevate the affected extremity.
5. Apply a warm compress to the site.
6. Mark the margins of the affected area and document the time of recognition.
7. Notify the receiving facility as early as possible so that definitive treatment can be prepared. Pharmacologic reversal is not carried under this protocol and is administered at the receiving facility.
8. Contact Medical Control.
9. Establish alternate vascular access and resume the infusion if still indicated.
10. Report as a sentinel event per Quality Improvement Program (Section 8-27).

Complications and Discontinuation

1. For new tachydysrhythmia, chest pain, or marked hypertension, reduce the rate or stop the infusion and contact Medical Control.
2. Do not stop a running infusion abruptly in a patient who remains dependent on it. Wean against the endpoint.
3. For any pump malfunction, stop the infusion, disconnect from the patient, and contact Medical Control.

Medical Control Contact Required

1. Before exceeding 10 mcg/min in an adult, or 1 mcg/kg/min in a pediatric patient.
2. Any suspected extravasation.
3. New tachydysrhythmia, chest pain, or marked hypertension during infusion.
4. Pump malfunction during infusion.

MCA Quality Improvement Performance Parameters

1. Review all cases in which an epinephrine infusion is initiated for compliance with this protocol. Review is conducted by the Professional Standards Review Organization per Quality Improvement Program (Section 8-27).
2. Review will address, at minimum:
    a. Time from patient contact to infusion initiation.
    b. Time from first push dose epinephrine to infusion initiation.
    c. Achievement of the titration endpoint within 10 minutes of initiation.
    d. Accuracy of documented concentration, rate, and titration.
    e. Discontinuation of push dose epinephrine following infusion initiation.
    f. Extravasation events and adherence to the extravasation response.
    g. Pump availability and function.
3. Any intervention where it is reasonable to believe harm to the patient may have occurred must be reported per Section 8-27, Sentinel Event Reporting.

Medication Protocols

Epinephrine (Section 9-23R)